10/9/26 Ex.no.7 Develop a Simple Calculator using XML-RPC.

Aim:

To develop a simple calculator using XML-RPC in python to perform basic arithmetic operations between a client and a server.

Procedure:-

* Create folder named XMLRPC-calculator.
* Create two files: Server.py and client.py
* Write the XML-RPC Server.py
* Write the Server using python Server.py.
* Write the client program in client.py
* Open another command prompt and run python client.py.
* Enter two numbers and Select addition, Subtraction, multiplication or division.

O/P

Result:

Thus, a Simple calculator was successfully developed using XML-RPC in python and arithmetic operations were performed between the client and server.

server.py - C:\Users\HDC0222020\Desktop\XMLRPC_Calculator\server.py (3.14.6)

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```
from xmlrpc.server import SimpleXMLRPCServer

server = SimpleXMLRPCServer(("localhost", 8000))

print("XML-RPC Calculator Server Running on Port 8000...")

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Division by zero is not allowed."
    return a / b

server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")

server.serve_forever()
```

client.py - C:\Users\HDC0222020\Desktop\XMLRPC_Calculator\client.py (3.14.6)

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```
import xmlrpc.client

server = xmlrpc.client.ServerProxy("http://localhost:8000")

a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

print("\nSelect Operation")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

choice = input("Enter your choice (1-4): ")

if choice == '1':
    print("Result =", server.add(a, b))
elif choice == '2':
    print("Result =", server.subtract(a, b))
elif choice == '3':
    print("Result =", server.multiply(a, b))
elif choice == '4':
    print("Result =", server.divide(a, b))
else:
    print("Invalid Choice")
```

C:\Windows\system32\cmd.exe - python server.py

```
C:\Users\HDC0222020\Desktop\XMLRPC_Calculator>python server.py
XML-RPC Calculator Server Running on Port 8000...
127.0.0.1 - - [10/Sep/2026 08:31:32] "POST /RPC2 HTTP/1.1" 200 -
```

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C:\Windows\system32\cmd.exe

```
C:\Users\HDC0222020\Desktop\XMLRPC_Calculator>

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op\XMLRPC_Calculator>python client.py
Enter First Number: 20
Enter Second Number: 10

Select Operation
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1
Result = 30

C:\Users\HDC0222020\Desktop\XMLRPC_Calculator>
```